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A hybrid deep learning framework for epileptic seizure prediction using scalp and intracranial EEG data☆

Izza Mujeeb Ahmad, Bisma Ashar, Shehryar Munir, Aamir Wali *, Muhammad Saif ul Islam

FAST School of Computing, National University of Computer and Emerging Science, Faisal Town, Lahore, Pakistan

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ABSTRACT

Epilepsy is a prevalent neurological disorder affecting over 50 million people globally, often impairing quality of life due to unpredictable and recurrent seizures. Despite advances in seizure prediction, challenges remain due to variability in EEG signals. EEG data exists in two primary modalities: non-invasive scalp EEG (sEEG) and invasive intracranial EEG (iEEG), and current deep learning models often exhibit limited generalizability across these modalities due to signal and structural differences. In this study, we propose a custom hybrid CNN-LSTM model optimized for scalp EEG, and assess its effectiveness across modalities by also evaluating it on invasive EEG datasets. We utilize the CHB-MIT (sEEG) and AES-Kaggle (iEEG) datasets employing a unified pipeline of preprocessing, signal transformation, and deep learning techniques. The model reported an accuracy of 98.84% on sEEG, using a 10-minute seizure anticipation window, outperforming conventional CNN, LSTM and several recent state-of-the-art models. On iEEG data, the hybrid model performed comparably well, achieving an impressive accuracy of 97.18%. The models are further validated using real-world scalp EEG data collected from the General Hospital Lahore, Pakistan. This additional validation demonstrates strong generalizability to clinical settings and confirms the practical applicability of the proposed approach. These findings not only highlight the effectiveness of the proposed approach for seizure prediction but also underscore its readiness for real-world deployment, potentially enabling reliable early warning systems to improve the lives of epilepsy patients.

1. Introduction

Globally, an estimated 5 million individuals are diagnosed with epilepsy annually, marking it as a significant neurological disorder with potentially life-threatening consequences. The documented risk of premature mortality in people with epilepsy is alarmingly high, reported to be up to three times greater than that of the general population (Cui et al., 2018). This escalating prevalence and the critical nature of epileptic seizures underscore the urgent need for proactive, preventive techniques rather than solely reactive interventions.

Electroencephalogram (EEG) is a widely used technique that records the brain's electrical activity through scalp-mounted electrodes (ul Islam and Farooq, 2017). EEG data can be acquired via non-invasive Scalp Electroencephalogram (sEEG) or invasive Intracranial Electroencephalogram (iEEG), with the latter offering higher spatial resolution and signal fidelity. The analysis of EEG data offers crucial insights that can substantially aid in the prediction of epileptic seizures. They can also serve as a valuable source of information for identifying patterns associated with the onset of epileptic seizures. Neurologists traditionally analyze EEG recordings to detect historical seizure patterns and assess seizure likelihood. However, the manual scrutiny of extensive EEG datasets to discern these critical preictal patterns is a labor-intensive and time-sensitive task.

Our work is motivated by the potential to assist neurologists through the development of an optimized computational solution that meets current state-of-the-art results. Recent studies have explored various deep learning architectures — such as attention mechanisms, temporal convolutional networks, and hybrid CNN-LSTM models — for automatic seizure classification and detection, often achieving high intra-dataset accuracy (Daoud et al., 2020; Liu et al., 2019; Varnosfaderani et al., 2021). However, many of these models are trained and validated on either scalp or intracranial EEG data exclusively, limiting their ability to generalize across modalities. Furthermore, while detection performance is often strong, predictive performance — especially

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* Corresponding author.
E-mail addresses: l201314@lhr.nu.edu.pk (I.M. Ahmad), l201386@lhr.nu.edu.pk (B. Ashar), l201302@lhr.nu.edu.pk (S. Munir), aamir.wali@nu.edu.pk (A. Wali), muhammad.saif@nu.edu.pk (M.S. ul Islam).

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with extended anticipation windows and real-world constraints — remains underexplored. These limitations highlight the need for robust, generalizable models capable of operating across heterogeneous EEG sources in realistic clinical environments.

From a clinical perspective, the proposed seizure prediction framework has the potential to serve as a component of a Clinical Decision Support System (CDSS) for epilepsy management. By providing accurate and timely identification of preictal states, such a system could alert healthcare professionals or caregivers in advance, enabling preventive interventions, personalized treatment adjustments, and improved patient safety. Integration into clinical workflows would require the model to operate in real-time, process continuous EEG streams, and deliver results through interfaces compatible with existing hospital information systems.

1.1. Problem statement

The core challenge lies in the accurate and timely prediction of epileptic seizures by analyzing EEG data. While existing research has made significant strides in seizure detection (identifying seizures as they occur), the anticipation of impending seizures remains a critical unmet need. Furthermore, many current approaches focus on either scalp or intracranial EEG data in isolation, potentially limiting the generalizability and robustness of the models. There is also a need for improved timeliness in prediction, enhanced sensitivity to minimize false negatives, and solutions that are computationally efficient for real-world deployment.

1.2. Objectives

In this study, we propose a hybrid CNN-LSTM model for the prediction of epileptic seizures. The primary objectives of this research are as follows:

- To design a hybrid deep learning framework that combines the spatial feature extraction capabilities of Convolutional Neural Network (CNN) with the temporal sequence modeling strength of Long Short-Term Memory (LSTM) for accurate seizure prediction.
- To rigorously evaluate the model's performance on both non-invasive scalp EEG data (*CHB-MIT dataset*) and invasive intracranial EEG data (*Kaggle American Epilepsy Society (AES) dataset*).
- To improve the accuracy and anticipation timing of seizure prediction, enabling reliable early warnings ahead of seizure onset.
- To explore strategies such as feature fusion for minimizing overfitting and improving the model's robustness for deployment in real-world healthcare settings.

1.3. Significance and impact

The successful development of an accurate and timely seizure prediction system holds significant potential to improve the quality of life for individuals living with epilepsy, their caregivers, and the healthcare professionals involved in their diagnosis and management. By providing early warnings before seizure onset, our proposed solution can enable timely medical intervention, potentially preventing injuries, reducing cognitive impairment, and alleviating the social isolation often associated with this condition. Furthermore, an efficient and robust prediction model can reduce the burden on neurologists by automating the initial analysis of EEG data, allowing them to focus on more complex cases and personalized treatment strategies.

1.4. Paper organization

The remainder of this paper is organized as follows: Section 2 provides a comprehensive review of the existing literature on epileptic seizure prediction using traditional and machine learning techniques, highlighting the research gaps that this work aims to address. Section 3 details the methodology employed, including the datasets used, preprocessing steps, feature extraction techniques, and the architecture of the proposed hybrid CNN-LSTM model. Section 4 presents and discusses the experimental results obtained on both scalp and intracranial EEG data, along with a comparison to state-of-the-art methods and an evaluation of seizure time anticipation. Finally, Section 5 concludes the paper and outlines potential directions for future research.

2. Literature review

Several traditional and machine learning-based methods have been proposed for epileptic seizure prediction. Slimen et al. (2020) employed a spike-based detection approach using EEG, triggering an alarm when spike counts exceeded a predefined threshold. Although simple and interpretable, this method was constrained by its exclusive reliance on scalp EEG and the absence of machine learning or deep learning integration. Yuan et al. (2018) introduced Bayesian Linear Discriminant Analysis to calculate diffusion distances between wavelet coefficients. However, its performance could be enhanced through a richer set of handcrafted features for better pattern discrimination.

Cui et al. (2018) applied EEG synchronization using a Bag-of-Words model and Extreme Learning Machine (ELM), suggesting that deeper learning models could yield better representation and predictive power. Similarly, Shiao et al. (2016) used Support Vector Machines on iEEG data from epileptic dogs, achieving 90% sensitivity. However, their subject-specific approach and the use of imbalanced datasets limited broader applicability. Usman et al. (2017) explored empirical mode decomposition with K-Nearest Neighbors (KNN), Naïve Bayes, and Support Vector Machine (SVM) classifiers, demonstrating effective feature extraction but lacking modern deep learning capabilities and being restricted to scalp EEG data.

With the advent of deep learning, Liu et al. (2019) proposed a Multiview CNN that leveraged both time and frequency EEG features. While effective, its high computational complexity posed scalability concerns. Daoud et al. (2020) developed an IoT-enabled deep learning model using DCNN for spatio-temporal feature classification. However, using data from only 8 out of 22 subjects reduced generalizability. Varnosfaderani et al. (2021) introduced an LSTM model with Swish activation, showing strong classification performance across patient-specific and patient-independent cases, though the lack of automated feature selection points to potential benefits from hybrid architectures.

Ryu and Joe (2021) combined DenseNet and LSTM in a hybrid architecture, surpassing 92% accuracy. Despite its strength, the model's dependence on pre-ictal segment assumptions and its exclusive use of scalp EEG data limit its universality. In Truong et al. (2019), a semi-supervised Generative Adversarial Networks (GAN) based method was explored, attaining a 75% accuracy. As a claimed first effort in applying GANs for seizure prediction, further refinement could boost its effectiveness. Yan et al. (2022) employed a three-tower transformer model but faced limitations in sensitivity and efficiency due to its simplistic architecture.

Early feature extraction techniques also contributed foundational insights. Fadzal et al. (2012) employed the Wavelet Transform, leveraging the properties of a mother wavelet, yet did not address computational efficiency. Hazarika et al. (1997) applied Short-Time Fourier Transform (STFT), noting its utility in temporal analysis but also indicating room for improvement in real-time processing contexts.

Recent studies have also explored graph-based deep learning and attention mechanisms to model inter-channel relationships in EEG signals explicitly. Graph Attention Networks (GATs) have shown promise in both seizure detection and broader epilepsy classification tasks by representing EEG channels as nodes and their functional or spatial relationships as edges. Mazurek et al. (2025) proposed a lightweight GAT-based framework designed for low-cost EEG hardware in low-resource settings, highlighting how learned attention weights enable visualization

Table 1
Review of ML-Based Epileptic Seizure Detection Studies.

Ref.	*Method*	*Result*	*Metric*
Slimen et al. (2020)	EEG Spike Detection	–	–
Yuan et al. (2018)	Wavelet + Bayesian LDA	85%	Sensitivity
Cui et al. (2018)	BoWav + ELM	85%	Sensitivity
Liu et al. (2019)	Multiview CNN	82%	Accuracy
Daoud et al. (2020)	DCNN (IoT-based)	–	–
Varnosfaderani et al. (2021)	LSTM + Swish Activation	–	–
Shiao et al. (2016)	SVM on Nonlinear Features	90%	Sensitivity
Bandarabadi et al. (2015)	Spectral Power + SVM	–	–
Usman et al. (2021)	EMD + k-NN/SVM/NB	–	–
Savadkoohi et al. (2020)	Wavelet + SVM/k-NN	92%	Accuracy
Ryu and Joe (2021)	DenseNet + LSTM	92%	Accuracy
Usman et al. (2017)	MAML-based Ensemble	91%	Sensitivity
Truong et al. (2019)	GAN-based Semi-Supervised	75%	F1-Score
Yan et al. (2022)	Transformer (Three-Tower)	91%	Accuracy
Hazarika et al. (1997)	STFT + Spectrogram	–	–
Fadzal et al. (2012)	Wavelet Transform	–	–
Liu et al. (2024)	Multi-dimensional Hybrid Bilinear CNN-LSTM	98.44%	Accuracy
Shawly et al. (2025)	MAFBN: Multi-Attention Forward and Backward Network	97.88%	Accuracy
Ghosh and Dey (2024)	CNN-BiLSTM-Attention with Multisynchrosqueezing Transform	99.70%	Accuracy
Patel (2024)	SeizureSight: 2D CNN-LSTM Hybrid	94.00%	Accuracy
Helgesen (2024)	Machine Learning-based Seizure Prediction	96.30%	Sensitivity
Fu et al. (2025)	WCE-MASE-PGCN	97.9%	Sensitivity

of critical channel-to-channel interactions and potential biomarkers. Building on this, Fu et al. (2025) introduced a Spatiotemporal Posterior Graph Convolutional Neural Network (WCE-MASE-PGCN) that integrates multi-head attention with a Squeeze-and-Excitation (SE) module. By combining graph convolutions with attention-driven feature recalibration, this approach enhances the model's ability to distinguish subtle preictal transitions while providing a sophisticated framework for mapping the spatiotemporal evolution of seizures.

Existing research spans from traditional signal processing methods to advanced deep learning architectures, each contributing unique insights into seizure prediction. While CNNs excel in extracting spatial and frequency-based features from EEG data, they often lack temporal sensitivity. Conversely, LSTMs effectively capture temporal dependencies but may underperform in spatial feature learning when used in isolation. The literature thus reveals a critical gap: the need for a unified framework that leverages the spatial learning strength of CNNs and the temporal sequence modeling capabilities of LSTMs. This motivates the development of a hybrid deep learning system designed to enhance predictive accuracy and generalizability across diverse EEG modalities. A comparative summary of the reviewed approaches is provided in Table 1.

3. Methodology and experimentation

This section outlines the proposed methodology for predicting epileptic seizures using EEG data. The approach begins with a detailed description of the datasets used, followed by a comprehensive data preprocessing pipeline that includes annotation, channel selection, and feature extraction. The classification task is framed as a binary classification problem, where the two target classes are: preictal, referring to the period leading up to a seizure (before onset), and ictal, representing the period during the seizure. The objective is to distinguish between these two states by identifying subtle changes in EEG patterns, thereby enabling early seizure prediction. This distinction is critical for real-world medical applications, as accurate preictal detection can provide patients and caregivers with timely warnings. The classification models evaluated include CNN, LSTM, and a hybrid CNN-LSTM architecture designed to capture both spatial and temporal features from EEG signals. CNNs have been used for all kinds of data such as images (Shahzad et al., 2025; Ather et al., 2024), audio/speech (Fawaz et al., 2021; Mahmood et al., 2024), videos (Hamza and Wali, 2023, 2025) and text (Shahzad and Wali, 2022; Shahid et al., 2024).

Table 2
Number of preictal and interictal segments in CHB-MIT and AES datasets.

Dataset	Preictal segments	Interictal segments
CHB-MIT	150	150
AES	211	211

3.1. Dataset

We employ two EEG datasets: the Children's Hospital Boston-Massachusetts Institute of Technology (CHB-MIT) Scalp EEG dataset (Goldberger and Amaral, 2010) and the American Epilepsy Society (AES) Seizure Prediction Challenge iEEG dataset (Cukierski, 2014). These datasets differ in recording modality and structure, which provides complementary perspectives on seizure activity. CHB-MIT comprises non-invasive scalp EEG recordings from 23 pediatric patients (aged 1.5–22 years) collected through multiple electrodes placed on the scalp. It contains long-term continuous recordings with varying seizure types, making it a widely used benchmark for seizure detection and prediction tasks.

In contrast, the AES dataset utilizes intracranial EEG (iEEG) recordings obtained via invasive electrode implantation directly within brain tissue, with labeled 10-minute segments categorized as interictal or preictal. The AES dataset includes long-duration intracranial EEG from 5 human and 2 canine subjects with labeled preictal and interictal segments. This dataset offers higher spatial resolution and signal-to-noise ratio, capturing localized brain activity that is critical for the precise detection of seizure onset. Using both datasets allows for a comprehensive evaluation of predictive models across different signal acquisition modalities, patient populations, and clinical settings. Specifications of the datasets are summarized in Table 2. Visualizations of channels for CHB-MIT and iEEG datasets are shown in Fig. 1 and Fig. 2, respectively.

While the iEEG dataset is well-suited for seizure prediction due to its pre-segmented and labeled structure, the CHB-MIT dataset was

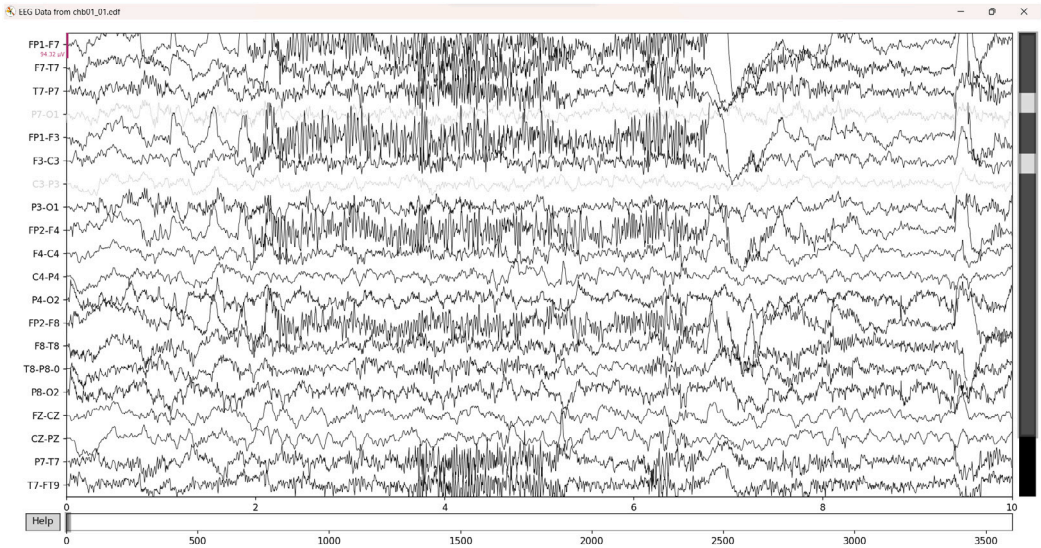

Fig. 1. Multiple channels for CHB-MIT scalp data.

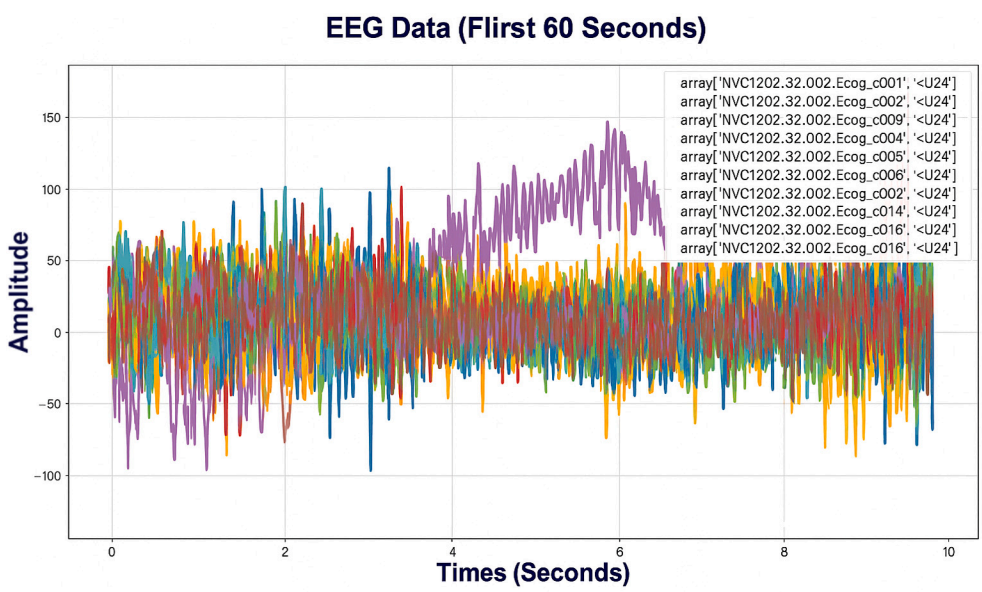

Fig. 2. Multiple channels for Kaggle iEEG data.

originally designed for seizure detection, not prediction. It contains no explicit labels for preictal or interictal states. Therefore, manual annotation was necessary to adapt it for predictive modeling. Additionally, most files in CHB-MIT span only around 1 h, limiting the possibility of defining long interictal windows far from seizure events within a single recording.

To ensure the proposed hybrid CNN-LSTM model generalizes beyond the datasets used for initial training, an additional external dataset was incorporated. This dataset consisted of scalp EEG recordings collected from 10 patients at General Hospital Lahore, Pakistan, including preictal and interictal segments. EEG data were collected using the KT88-1032 multichannel digital EEG system (supporting 18–32 channels), with electrodes positioned in accordance with the International 10–20 system. All recordings were preprocessed using the same pipeline as the CHB-MIT and AES datasets to maintain consistency. This external dataset was used exclusively for final testing and was not involved in any stage of model training, validation, or hyperparameter optimization.

3.2. Data preprocessing

3.2.1. Data annotation and sampling

To adapt the CHB-MIT dataset for seizure prediction, we implemented a manual labeling strategy to determine the preictal periods. This process included neurologist input and visualization of EEG traces to manually identify intervals of 10, 20, and 30 min immediately preceding the seizure onset. These durations were chosen to evaluate how early the model can predict an upcoming seizure. Experiments were conducted on all three sets of data, and the most optimal time of 30 min was selected for experimentation based on the models' performance.

Due to the limited recording duration (often 1 h), identifying interictal segments several hours away from a seizure was not feasible. Instead, we selected interictal windows that were at least 40 min away from any seizure onset within the same or different seizure-free recording files from the same patient. This helped ensure the selected interictal data were not contaminated by nearby preictal states.

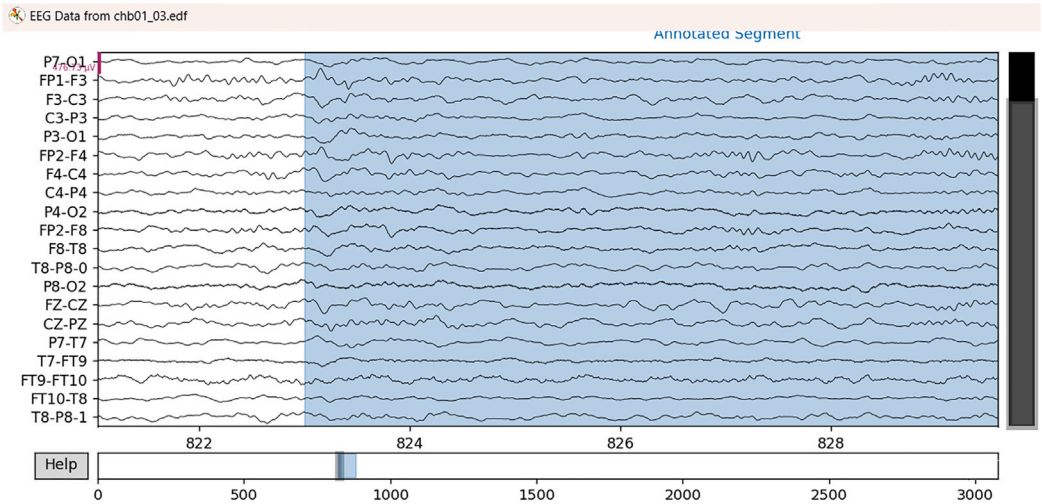

Fig. 3. MNE library used to visualize and annotate seizure data in the EDF file.

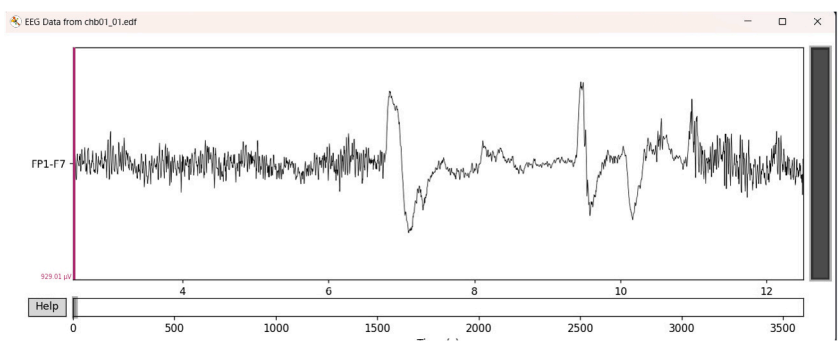

Fig. 4. Selection of the FP1-F7 channel from all the electrode channels using a script.

MNE is a Python library for neurophysiological data that effectively reads and processes EEG data. This library was used to visualize and annotate EEG EDF files. This involved using a translucent color overlay to highlight the segments where seizures were detected. This whole step helped to obtain a detailed visual analysis of seizure and non-seizure data and separate them. The MNE library and how it visualizes data are shown in Fig. 3.

3.2.2. Class imbalance

Class imbalance, i.e., preictal class being underrepresented in the dataset, can adversely affect model performance. To address this concern, we worked with an expert neurologist to select only a subset of normal (interictal) data that was as far away as possible from the ictal region. The remaining normal data of 30 min duration each were randomly selected from files of patients without seizures, matching the count of seizure samples. This ensured equal representation of both classes.

3.2.3. Channel selection

The CHB-MIT (sEEG) and AES-Kaggle (iEEG) datasets differ in EEG acquisition setups, with variations in electrode placement, number of channels, and signal morphology due to scalp vs. intracranial recording. To account for these differences, we processed each dataset independently and developed dataset-specific pipelines.

In the CHB-MIT dataset, patient EEGs were recorded with electrode channels on their scalp, and these channels in some cases differed between patients. Therefore, EEG channels that were common to all patients were selected. Furthermore, erroneous EEG channels were also excluded to mitigate the impact of faulty channels on the prediction model. For the CHB-MIT dataset, the channels '-', '–0', '–1', '–2', '–3', '–4', and '–5' were excluded as their purpose was to include metadata. As a result, 16 intersecting channels across all recordings were chosen: 'FP1-F7', 'F7-T7', 'T7-P7', 'P7-O1', 'FP1-F3', 'F3-C3', 'C3-P3', 'P3-O1', 'FP2-F4', 'F4-C4', 'C4-P4', 'P4-O2', 'FP2-F8', 'F8-T8', 'T8-P8', and 'P8-O2', 'FZ-CZ', 'CZ-PZ'. Some of these channels are named differently for Patient 12. Fig. 4 shows how one of these channels was separated from all other channels using scripts.

The AES-Kaggle dataset, on the other hand, contains uniform electrode configurations and pre-segmented labeled intervals. As a result, no cross-modal channel alignment was required.

3.2.4. Butterworth bandpass filter

A Butterworth bandpass filter was applied to the EEG data to reduce noise and preserve signal integrity. This filter was chosen for its maximally flat passband, which ensures minimal signal distortion, critical for detecting subtle spectral and temporal features in EEG analysis (Craik et al., 2019). The selected frequency range, 0.5–40 Hz, is standard in EEG preprocessing (Acharya et al., 2013). Frequencies below 0.5 Hz were excluded to eliminate slow drifts and baseline wander caused by sweat or movement artifacts. Frequencies above 40 Hz were removed to suppress muscle artifacts and power-line interference (50/60 Hz), which do not contribute meaningfully to seizure prediction and can degrade model performance. The Butterworth filter was preferred over alternatives like Chebyshev or Elliptic filters because it avoids ripples in both the passband and stopband, maintaining a smoother frequency response. Although Chebyshev and Elliptic filters offer steeper roll-off, they often introduce phase and amplitude distortions, which can compromise the fidelity of EEG signals (Widmann et al., 2015). Fig. 5 represents the filtered signal after applying the BFB filter. Note that the Butterworth bandpass filter was selected not only

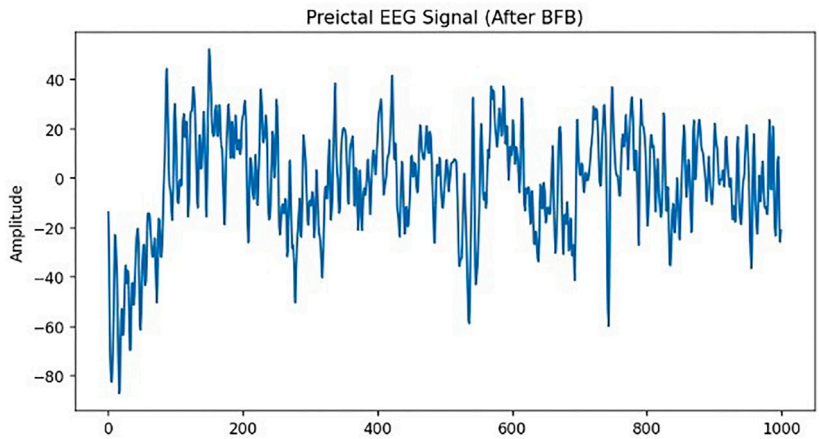

Fig. 5. Applying Butterworth filter on iEEG data.

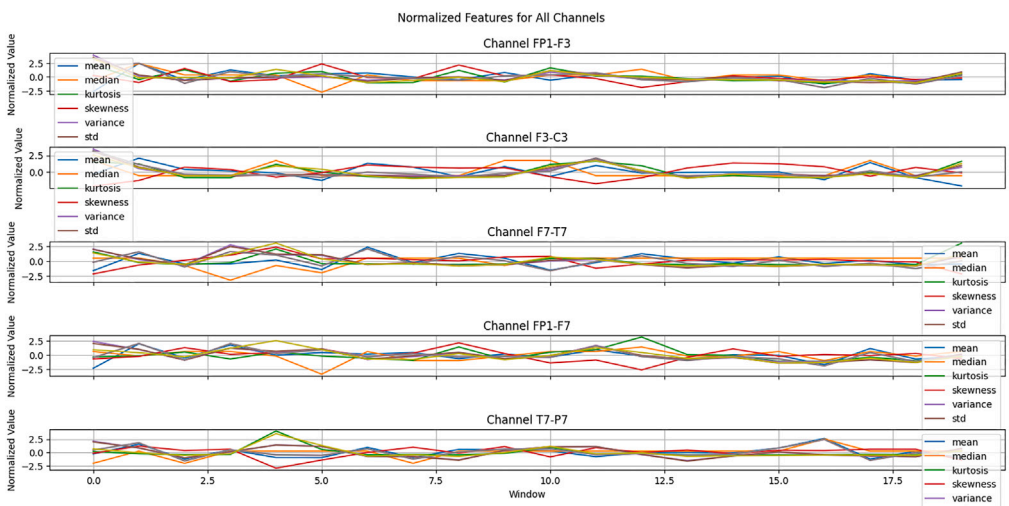

Fig. 6. Features extracted for various channels in iEEG data.

for its favorable frequency characteristics but also based on empirical evidence — unfiltered EEG data yielded classification accuracies below 60%, indicating that preprocessing was essential to enhance signal quality and model performance.

3.3. Feature extraction

3.3.1. Handcrafted feature extraction

Handcrafted features were generated from 20-second overlapping windows with a 50% overlap of EEG data. The final set of features included statistical measures, including mean, median, kurtosis, skewness, variance, standard deviation, power, root mean square (RMS), zero-crossing rate (ZCR), and signal envelope. Fig. 6 presents the handcrafted features extracted across multiple channels in the iEEG dataset, demonstrating spatial variability and feature diversity relevant to seizure prediction.

3.3.2. Short-time fourier transform

The Short-Time Fourier Transform (STFT) was utilized to extract frequency domain features from the EEG data segments. This technique provided a time-frequency representation of the EEG signals, which is valuable for identifying patterns associated with seizure events. From this representation, spectrogram images, which visually represent the frequency content of the EEG signals over time, were generated. Fig. 7 shows a spectrogram extracted from a segment of channel 1. Various window lengths and overlaps were experimented with during the feature extraction process. The EEG windows, once transformed into spectrograms using Short-Time Fourier Transform, are fed into the CNN model to extract a feature vector.

3.4. Hardware and software requirements

The experiments were conducted on a system equipped with the following hardware components: an AMD Ryzen 7 5800H CPU and a GeForce RTX Ti graphics processing unit (GPU), along with 64 GB of RAM.

On the software side, the development environment was configured using Python 3.10 as the primary programming language. For deep learning model implementation, we utilized TensorFlow 2.15.0 coupled with Keras 3.1.1 as the high-level neural network API. This combination allowed for efficient model design and execution and leveraged the GPU for accelerated training. MATLAB was immensely useful for visualizing signal data in various forms.

3.5. Model architecture

In this study, we propose a CNN-LSTM model, where CNN functions as a feature extractor while LSTM is used for time series classification. Given that the proposed architecture integrates two distinct deep learning models—CNN and LSTM—we also evaluate each model independently as a baseline for comparative analysis. The architectural details of the individual CNN and LSTM models, along with the combined CNN-LSTM framework, are presented in this section.

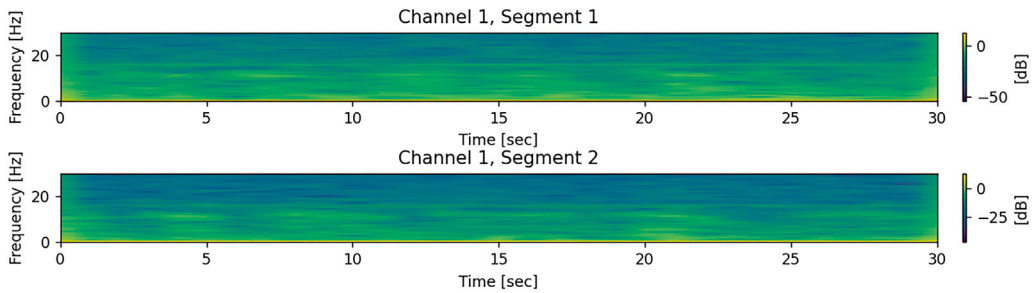

Fig. 7. sample spectrograms extracted using STFT from channel 1.

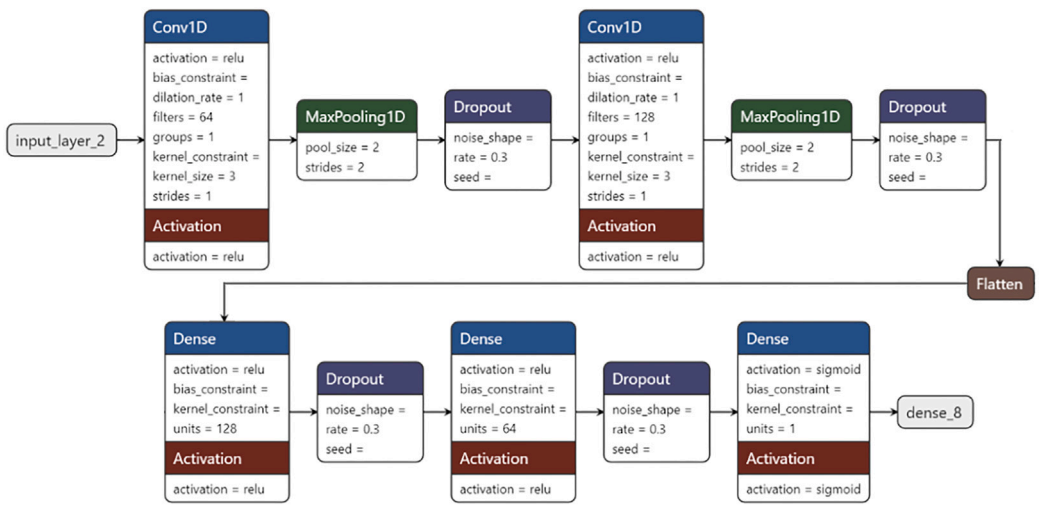

Fig. 8. CNN Model Architecture.

3.5.1. Convolutional neural network (CNN)

A Convolutional Neural Network model is employed for spectrogram analysis. Transfer learning with ResNet, MobileNet, and EfficientNet-B3, as well as a custom CNN model trained from scratch, was explored to enhance model performance. The custom model performed most optimally as discussed in the results section.

A custom CNN model has been developed that processes 2D spectrograms with an input shape of 129 × 92. The architecture includes two Conv1D layers with 64 and 128 filters, respectively. Both of the layers use a kernel size of 3 and ReLU activation. Each convolutional layer is followed by a MaxPooling1D layer with a pool size of 2 and a Dropout layer having a dropout rate of 0.3, specifically for regularization. The model ends with a Flatten layer, followed by Dense layers with 128 and 64 units. Each of these layers is accompanied by a dropout for optimal regularization. The final output layer uses sigmoid activation for binary classification, i.e., the preictal stage or the interictal stage. The architecture of the custom CNN is provided in Fig. 8.

3.5.2. Long short-term memory (LSTM)

The analysis of the problem led to the understanding that for a 10-minute segment, each 20-second spectrogram image is treated separately, and the context for the sequence is not understood. Therefore, RNNs are crucial to solving this problem more efficiently. A Long Short-Term Memory network is designed to efficiently capture the temporal dependencies in the combined feature vector obtained from the CNN and handcrafted features. LSTMs are crucial for sequence prediction problems due to their ability to maintain long-term dependencies and, most importantly, reduce the vanishing gradient problem common in traditional Recurrent Neural Networks (RNNs). The LSTM component of the model comprises two layers. The first layer consists of 100 units, which enables the network to learn complex temporal patterns in the EEG data. This is followed by a dropout rate of 0.2. The same alternating structure of layers is followed next. A Dense layer with a single unit and sigmoid activation function is used for the final binary classification, distinguishing between interictal and preictal EEG states. The architecture of the custom LSTM network is provided in Fig. 9.

3.5.3. Hybrid CNN-LSTM model

In the proposed hybrid architecture, we integrate both handcrafted and deep-learned features to enhance seizure prediction accuracy. First, spectrograms are generated from EEG signal segments to capture time-frequency information. These spectrograms are then fed into a custom CNN, which extracts deep feature representations. Simultaneously, handcrafted features are derived from the same EEG segments to capture statistical and signal-based characteristics.

The deep CNN features are concatenated with the handcrafted features, forming a comprehensive feature matrix that encapsulates both learned and domain-specific insights. This hybrid feature representation serves as the input to the LSTM model, which is designed to capture temporal dependencies in the sequential data. To ensure proper label alignment, class labels are matched accordingly with each feature matrix.

This hybrid approach leverages the CNN's strength in extracting spatial and frequency-domain patterns, alongside the LSTM's capability to model temporal dynamics, ultimately leading to a robust and generalizable prediction framework.

Methodological contribution. While CNN-LSTM combinations have been explored in prior studies, our proposed framework introduces three important innovations: (i) the use of a custom, lightweight CNN architecture tailored for spectrogram-based seizure prediction; (ii) the integration of handcrafted EEG signal features (e.g., skewness, kurtosis,

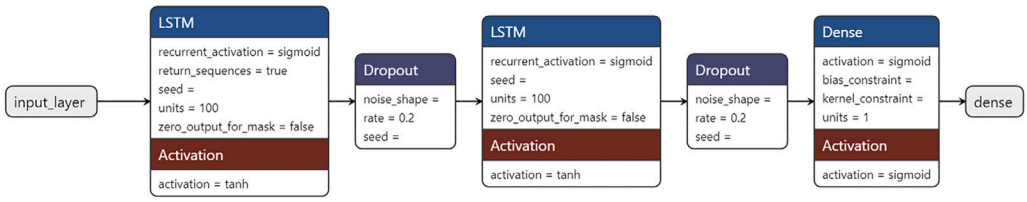

Fig. 9. LSTM Model Architecture.

Table 3
Validation accuracy on CHB-MIT and testing accuracy on PINS-sEEG using the Custom CNN-LSTM model. Anticipation time for all experiments: 600 s.

Feature extraction	*Validation Acc. (% ± SD)*	*Testing Acc. (%)*
Handcrafted statistical features (mean, variance, skewness, standard deviation, kurtosis, entropy)	94.96 ± 1.10	84.50
Short-Time Fourier Transform (STFT)	97.52 ± 0.78	87.22
Handcrafted + STFT	98.84 ± 0.54	89.15

ZCR), chosen in consultation with clinical experts, with CNN-extracted features to form a hybrid input; and (iii) the demonstration of generalization across scalp and intracranial EEG modalities using a unified training and evaluation pipeline. This design not only improves predictive accuracy but also supports real-time feasibility, enhancing the model's translational value for clinical settings.

3.6. Training procedure

We used subject-independent five-fold cross-validation to ensure reliable performance estimation and prevent data leakage. For the CHB-MIT dataset, EEG recordings from distinct patients were divided into five non-overlapping subsets, ensuring that data from any single subject appeared in only one fold. In each iteration, four folds were used for training and one for testing. This subject/session-wise partitioning prevents overlap between training and test data and avoids the risk of learning subject-specific patterns. The process was repeated five times, and the final accuracy was computed as the average of the five individual test accuracies, providing a robust estimate of the model's generalization to unseen subjects.

The CNN model was trained for 100 epochs using the Adam optimizer with an initial learning rate of 0.001 and binary cross-entropy as the loss function. A learning rate scheduler was applied to exponentially decay the learning rate by a factor of 0.9 per epoch. Additionally, a ReduceLROnPlateau callback was used to adjust the learning rate dynamically. If the validation loss did not improve for five consecutive epochs, the learning rate was reduced by a factor of 0.2, with a minimum threshold of 0.001. To prevent overfitting, early stopping was employed with a patience of five epochs. The model was trained with a batch size of 32. These training parameters were selected to ensure stable convergence and generalization across subjects and are reported here to support reproducibility. The primary evaluation metric was accuracy, with particular emphasis on the model's ability to increase anticipation time for seizure onset.

It is important to note that the CHB-MIT and AES-Kaggle datasets were used independently. For each dataset, the model was trained and evaluated using 5-fold cross-validation, and no cross-dataset training or testing was performed. This design allows for assessing the model's ability to generalize within each EEG modality, without introducing bias from cross-domain transfer.

4. Results and discussion

This section presents the experimental evaluation of the proposed CNN-LSTM model across three stages: performance on scalp EEG (sEEG), generalization to intracranial EEG (iEEG), and robustness under varying seizure anticipation times and real-world data. In all experiments, a Butterworth bandpass filter was applied during preprocessing to remove noise and retain relevant EEG frequency bands.

4.1. Results on scalp EEG (sEEG) data

We first evaluated the proposed model on the CHB-MIT scalp EEG dataset to assess its core performance on non-invasive data. As shown in Table 3, the CNN-LSTM model achieved its highest accuracy of 98.84% using a combination of handcrafted statistical features and Short-Time Fourier Transform (STFT). This demonstrates the effectiveness of feature fusion for seizure prediction on scalp data and confirms the model's ability to handle lower-resolution scalp signals. The recall is 0.99, the precision is 0.9803, and the FPR is 1.16%. The same model was then tested on an unseen dataset (PINS-sEEG from General Hospital Lahore), and the proposed method achieved an accuracy of 89.15% indicating the strong generalization ability of the model.

4.2. Generalization on invasive EEG (iEEG) data

We next assessed the model's generalizability to invasive EEG data using the AES-Kaggle dataset. While we use the term "generalization" in this section, it refers to the model's robustness when applied to a different EEG modality (iEEG) using the same training pipeline—not to training on one dataset and testing on the other. Each dataset was treated as an independent validation scenario.

Table 4 presents the performance of all models on the AES iEEG dataset. Since the proposed architecture integrates two distinct deep learning models, CNN and LSTM, we first conducted experiments using each of these models independently. ResNet achieved the lowest performance across both feature types, reaching 61.18% with handcrafted features and 55.61% with STFT features. In contrast, the custom CNN trained from scratch achieved 83.97% with handcrafted features and 85.61% with STFT. This shows that a task-specific shallow CNN extracts spatial patterns from EEG signals more effectively than the deeper ResNet architecture. The LSTM model further improved performance, obtaining 88.29% with handcrafted features and 89.47% with STFT. This improvement reflects the importance of temporal modeling in EEG analysis.

The proposed hybrid CNN-LSTM model achieved the highest accuracy across all configurations. It reached 94.12% with handcrafted features and 96.36% with STFT. When both handcrafted and STFT features were combined, accuracy increased to 97.18%, which is the best overall result.

The false alarm rate, also known as the false positive rate is also a very important indicator for the detection and prediction of epilepsy. Fig. 10 shows for the binary classification of preictal and interictal EEG segments for the iEEG data using our custom CNN-LSTM model. In this context, the positive class corresponds to preictal segments (i.e., intervals preceding seizure onset), while the negative class corresponds to interictal segments (i.e., seizure-free periods between events). The

Table 4
Accuracy (% ± SD) of different models on AES iEEG dataset. Anticipation time for all experiments: 600 s.

Feature Extraction	*ResNet*	*Custom CNN*	*LSTM*	*CNN-LSTM*
Handcrafted features	61.18 ± 1.15	83.97 ± 0.95	88.29 ± 0.88	94.12 ± 1.05
Short-Time Fourier Transform (STFT)	55.61 ± 1.20	85.61 ± 0.72	89.47 ± 0.72	96.36 ± 0.82
Handcrafted + STFT	65.23 ± 0.50	87.16 ± 0.82	91.48 ± 0.91	97.18 ± 0.65

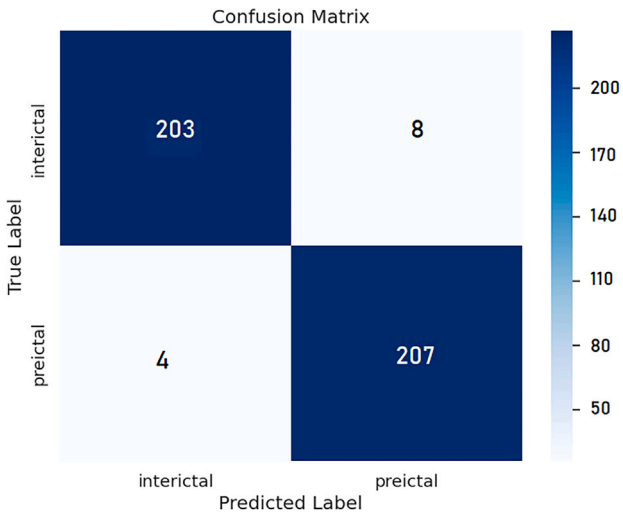

Fig. 10. Confusion matrix representing model evaluation for iEEG data.

model records a recall of 0.98 and a precision of 0.96. The false positive rate is 0.037.

Fig. 10 presents the confusion matrix. The model achieved **207 true positives** and **203 true negatives**, with only **8 false positives** and **4 false negatives**. This translates to a recall of **99.33%**, precision of **96.28%**, false positive rate (FPR) of **3.79%**, and false negative rate (FNR) of **1.90%**, yielding an overall accuracy of **97.16%**. The low FPR indicates that the model generates few false seizure warnings, which is critical for clinical applicability, while the high recall demonstrates its effectiveness in detecting imminent seizures with minimal missed events.

4.3. Seizure time anticipation

The effectiveness of the proposed CNN-LSTM model in early seizure prediction was evaluated using varying anticipation windows of 5 and 15 min, as summarized in Table 5. On the CHB-MIT (sEEG) dataset, the model achieved a peak accuracy of 99.24% with a 5-minute anticipation window. As the anticipation window extended to 15 min, accuracy slightly decreased to 97.35%, indicating a trade-off between early prediction and predictive performance. Similarly, on the AES (iEEG) dataset, the model maintained strong performance, achieving 98.67% accuracy at 5 min and 96.13% at 15 min.

These results show that shorter anticipation windows lead to higher prediction accuracy, likely due to stronger preictal signal characteristics closer to the onset. Longer windows introduce greater variability and noise, making reliable prediction more difficult. Still, the model demonstrates stable performance across both sEEG and iEEG datasets, highlighting its adaptability to different electrode types and recording setups. This consistency suggests that the CNN-LSTM framework can serve as a strong candidate for early warning systems in clinical or home-based EEG monitoring applications, especially where rapid and accurate prediction is critical.

Table 5
Prediction using a combination of STFT and handcrafted features with different anticipation times of seizure.

Data	*Accuracy*	*Time (s)*
sEEG	99.24%	300
sEEG	97.35%	900
AES iEEG	98.67%	300
AES iEEG	96.13%	900

4.4. Testing on local sEEG recordings from Pakistan

To further validate the model in real-world clinical settings, scalp EEG recordings were obtained from a patient at Punjab Institute of Neurosciences (PINS), located at Lahore General Hospital, Pakistan. This dataset included a documented seizure event. The dataset was taken after IRB approval. It consisted of 5 segments with labeled preictal and interictal segments. Each segment was 20 min in duration.

Experimental analysis was conducted on both the preictal and interictal states using the proposed pre-trained model. The model was pre-trained on the sEEG dataset. Despite hardware differences (KT88-1032 device with 18–32 channels), preprocessing and channel realignment enabled model compatibility. The model achieved an accuracy of 89.15% on this dataset, demonstrating strong practical applicability under previously unseen conditions. The high precision (90.12%) indicates a low false alarm rate, while the recall of 88.47% reflects the model's effectiveness in correctly identifying seizure events. Furthermore, the specificity of 89.24% and a relatively low false positive rate of 10.84% confirm the model's robustness in distinguishing non-seizure activity. Collectively, these results highlight the reliability and generalization capability of the proposed approach for real-world EEG-based seizure prediction.

4.5. Comparison with state-of-the-art

To benchmark the proposed CNN-LSTM model, we compared its performance with recent state-of-the-art approaches for epileptic seizure detection using the CHB-MIT scalp EEG dataset. Table 6 summarizes the comparative results based on reported accuracies.

The proposed hybrid CNN-LSTM model achieved a classification accuracy of 98.84%, exceeding the performance of several recent models, including the hybrid CNN-LSTM by Liu et al. (2024) and the MAFBN architecture by Shawly et al. (2025). Although Ghosh and Dey's attention-augmented model reached a higher accuracy (99.70%), it relies on a computationally intensive architecture involving complex time-frequency transformations and deep recurrent neural networks. In contrast, our approach offers a much shallower, custom CNN-LSTM model that not only generalizes effectively across EEG modalities, including invasive iEEG, but is also well-suited for real-world clinical deployment due to its lower computational overhead.

4.6. Clinical integration and real-world utility

Beyond strong predictive performance, the proposed model aligns with the broader goal of clinical deployment by offering practical utility for integration into CDSS. In real-world scenarios, timely and accurate seizure prediction can enable clinicians to take preemptive measures, adjust medication, or alert caregivers. Our unified preprocessing and prediction pipeline is lightweight enough for real-time implementation

Table 6
Comparison of the proposed CNN-LSTM with State-of-the-Art methods using the CHB-MIT dataset.

Ref.	*Year*	*Model*	*Accuracy*
Liu et al. (2024)	2024	Multi-dimensional Hybrid Bilinear CNN-LSTM	98.44
Shawly et al. (2025)	2025	MAFBN: Multi-Attention Forward and Backward Network	97.88
Ghosh and Dey (2024)	2024	CNN-BiLSTM-Attention with Multisynchrosqueezing Transform	99.70
Patel (2024)	2024	SeizureSight: 2D CNN-LSTM Hybrid	94.00
Helgesen (2024)	2024	Machine Learning-based Seizure Prediction	96.30
This Study	2025	Custom CNN-LSTM (Handcrafted + STFT)	98.84

in hospital monitoring systems or wearable EEG devices. Moreover, by validating the model on clinical EEG data from General Hospital Lahore, we demonstrate its robustness in non-controlled, real-world environments. This paves the way for its integration into existing clinical workflows as a seizure alert module within a CDSS framework. Such integration can assist medical professionals in monitoring high-risk patients, reducing false alarms, and improving treatment personalization through data-driven insights.

4.7. Limitations and challenges

The persistent high imbalance in the interictal-to-ictal ratio remains a challenge that may impact the model's generalization in real-time systems. Furthermore, while a subject-independent training approach was explored, the significant inter-subject variability in seizure patterns indicates that personalized models could provide further performance enhancements, an aspect not fully addressed in this work. Although local EEG testing in Pakistan demonstrated a promising 89.15% accuracy, practical implementation revealed challenges with hardware compatibility and signal variation, highlighting the need for more robust preprocessing and model adaptation techniques. Lastly, even with the model's computationally efficient design, its seamless deployment on low-power wearable devices for continuous real-

5. Conclusion and future work

This study presents a robust and clinically relevant framework for epileptic seizure prediction using deep learning across both non-invasive scalp EEG (sEEG) and invasive intracranial EEG (iEEG) modalities. By leveraging the CHB-MIT and AES-Kaggle datasets, we designed and evaluated a custom CNN, LSTM, and a hybrid CNN-LSTM architecture. Among these, the proposed hybrid model demonstrated superior performance, achieving 98.84% accuracy on sEEG and 97.18% on iEEG with a 10-minute anticipation window.

A key strength of our approach lies in its ability to combine spatial and temporal representations through handcrafted features and STFT-based spectral analysis, enabling accurate seizure prediction with extended lead times. Unlike more computationally intensive models, our framework remains lightweight, generalizes well across modalities, and supports real-time feasibility. This was further confirmed through validation on real-world scalp EEG data collected from General Hospital Lahore, where the model maintained high performance under non-ideal conditions.

Overall, the findings validate our four core design objectives: improved predictive accuracy, generalization across modalities, reduced complexity, and early seizure anticipation. The results highlight the potential of the proposed model as a practical foundation for real-time, patient-centered early warning systems that can enhance proactive care and significantly improve the quality of life for individuals with epilepsy.

Future work will focus on expanding the datasets to include more diverse subjects and seizure types to enhance model robustness and reduce bias. Additionally, deploying the model on edge devices or mobile platforms will be explored for real-time prediction. We also aim to investigate personalized models through transfer learning or federated learning, enabling tailored predictions while preserving patient privacy. While the present study focuses on maximizing predictive performance and deployment feasibility using a hybrid CNN–LSTM architecture, future work will explore the integration of attention-based and graph neural network approaches to enhance model explainability. Finally, integration with wearable EEG hardware and collaboration with clinical institutions will be essential for transitioning this research from the lab to the bedside.

CRediT authorship contribution statement

Izza Mujeeb Ahmad: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Bisma Ashar:** Writing – original draft, Validation, Methodology. **Shehryar Munir:** Writing – original draft, Methodology, Data curation, Conceptualization. **Aamir Wali:** Writing – review & editing, Writing – original draft, Validation, Supervision. **Muhammad Saif ul Islam:** Writing – review & editing, Supervision.

Compliance with ethical standards

This statement is to certify that the author list is correct. The Authors also confirm that this research has not been published previously and that it is not under consideration for publication elsewhere. On behalf of all Co-Authors, the Corresponding Author shall bear full responsibility for the submission. There is no conflict of interest. This research did not involve any human participants and/or animals.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

sEEG dataset was manually annotated. The dataset is available at: https://physionet.org/content/chbmit/1.0.0/.

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Izza has a BS in Software Engineering. She was a research assistants in the DataGen Lab, working on EEG data acquisition and analysis.

Bisma has a BS in Software Engineering. She was a research assistants in the DataGen Lab, working on EEG data acquisition and analysis.

Shehryar has a BS in Software Engineering. He is a research assistants in the DataGen Lab, working on EEG data acquisition and analysis.

Dr. Aamir Wali is Professor and Head of Data Science at FAST NUCES and Director of the DataGen Lab, conducting research on EEG data acquisition and analysis.

Mr. Saif is a Lecturer in Data Science at FAST NUCES. He holds a Master's in Data Science and specializes in time series analysis, and provided his contributions in this domain.